

EIA & Environmental Legislation PYQs

Last 3 Years PYQ Solution (2022-2024)

2 SEPTEMBER 2024

Which of the following protocols are linked to convention on Biological Diversity?

A. Cartagena

B. Kigali

C. Nagoya

D. Rotterdam

E. UN-REDD

Choose the most appropriate answer from the options given below:

1. A and C Only

2. B and D Only

3. C and E Only

4. A and D Only

ANSWER: A

The Cartagena Protocol on Biosafety and the Nagoya Protocol on Access and Benefit-Sharing are linked to the Convention on Biological Diversity.

So, the most appropriate answer is:

1. A and C Only

Let us delve deeper into these protocols:

1. Cartagena Protocol on Biosafety

The Cartagena Protocol on Biosafety is an international agreement which aims to ensure the safe handling, transport, and use of living modified organisms (LMOs) resulting from modern biotechnology. It focuses particularly on movements across borders. Adopted in 2000 as a supplementary agreement to the Convention on Biological Diversity (CBD), it emphasizes the precautionary approach. Its key feature is the Advance Informed Agreement (AIA) procedure, which requires exporters to provide detailed information to importing countries, ensuring they can make informed decisions.

2. Nagoya Protocol on Access and Benefit-Sharing

The Nagoya Protocol, adopted in 2010, is another supplementary agreement to the CBD. It deals with access to genetic resources and the fair and equitable sharing of benefits arising from their utilization. This protocol aims to create greater legal certainty and transparency for both providers and users of genetic resources. It establishes more predictable conditions for access to genetic resources and helps to ensure benefit-sharing when genetic resources leave the contracting party providing them.

Why Not the Other Options?

- **Kigali Amendment:** This relates to the Montreal Protocol on Substances that Deplete the Ozone Layer, not the CBD.
- **Rotterdam Convention:** It deals with the Prior Informed Consent (PIC) procedure for certain hazardous chemicals and pesticides in international trade.
- **UN-REDD Programme:** This is the United Nations Collaborative Programme on Reducing Emissions from Deforestation and Forest Degradation in developing countries, unrelated to the CBD.

So, when it comes to protocols linked to the Convention on Biological Diversity, the correct answers are Cartagena and Nagoya.

"The formalized systematic and comprehensive process of evaluating the environment impacts of a policy, programme plan as its alternatives as well as using the findings in publicly accountable decision making" refers to

1. Comprehensive Impact Assessment (CIA)
2. Environmental Impact Assessment (EIA)
3. Interpreted Impact Assessment
4. Strategic Environmental Assessment

ANSWER: D

The process you're referring to is called **Strategic Environmental Assessment (SEA)**. This is a comprehensive and systematic method used to evaluate the environmental impacts of proposed policies, plans, and programs, as well as their alternatives. It aims to ensure that environmental considerations are integrated into decision-making processes in a way that is publicly accountable.

So, the correct answer is:

4. Strategic Environmental Assessment

Let us delve deeper into the concept of **Strategic Environmental Assessment (SEA)**.

Strategic Environmental Assessment (SEA)

Definition: SEA is a formalized, systematic, and comprehensive process used to evaluate the environmental impacts of proposed policies, plans, and programs, as well as their alternatives. The primary goal is to integrate environmental considerations into higher-level decision-making processes and ensure that these considerations are publicly accountable.

Key Elements of SEA

1. **Scope and Objectives**: SEA typically focuses on a broad range of environmental issues at a policy, plan, or program level, rather than individual projects.
2. **Alternatives Evaluation**: One of the critical components of SEA is the evaluation of different alternatives. This involves considering various options and their potential environmental impacts, helping decision-makers choose the most sustainable and least harmful alternative.
3. **Public Participation**: SEA involves stakeholders and the public in the decision-making process. This ensures that the views and concerns of those affected by the policy, plan, or program are considered.
4. **Integrated Approach**: Unlike Environmental Impact Assessment (EIA), which is usually project-specific, SEA is more strategic and integrates environmental, social, and economic considerations into policy, plan, and program development.
5. **Decision-Making**: The findings from the SEA process are used to inform and improve decision-making. This ensures that the decisions are environmentally sound and socially equitable.

Benefits of SEA

- **Sustainability**: By considering environmental impacts at the strategic level, SEA promotes sustainable development.
- **Informed Decisions**: It provides a systematic approach to identifying environmental impacts early in the planning process, leading to more informed decisions.
- **Transparency**: Involving the public and stakeholders enhances transparency and accountability in the decision-making process.
- **Conflict Prevention**: Identifying potential environmental issues early can help prevent conflicts and reduce the risk of costly mistakes.

Comparison with Environmental Impact Assessment (EIA)

While SEA and EIA both aim to assess environmental impacts, they differ in scope and application:

- **SEA** is applied to policies, plans, and programs at a strategic level, focusing on broader issues and alternatives.
- **EIA** is applied to specific projects, such as the construction of a dam or highway, assessing the direct environmental impacts of that project.

Conclusion

Strategic Environmental Assessment is a crucial tool for ensuring that environmental considerations are integrated into the early stages of policy, plan, and program development. By doing so, it promotes sustainable development, transparency, and informed decision-making.

Arrange the following steps in order of execution of Environmental Impact Assessment (EIA) process.

A. Analysis

B. Decision Making

C. EIA Reporting

D. Mitigation

E. Review

Choose the most appropriate answer from the options given below:

1. E. D. A, B, C

2. C. A, D, B. E

3. A, D, C, E, B

4. A, C, B, E, D

ANSWER: C

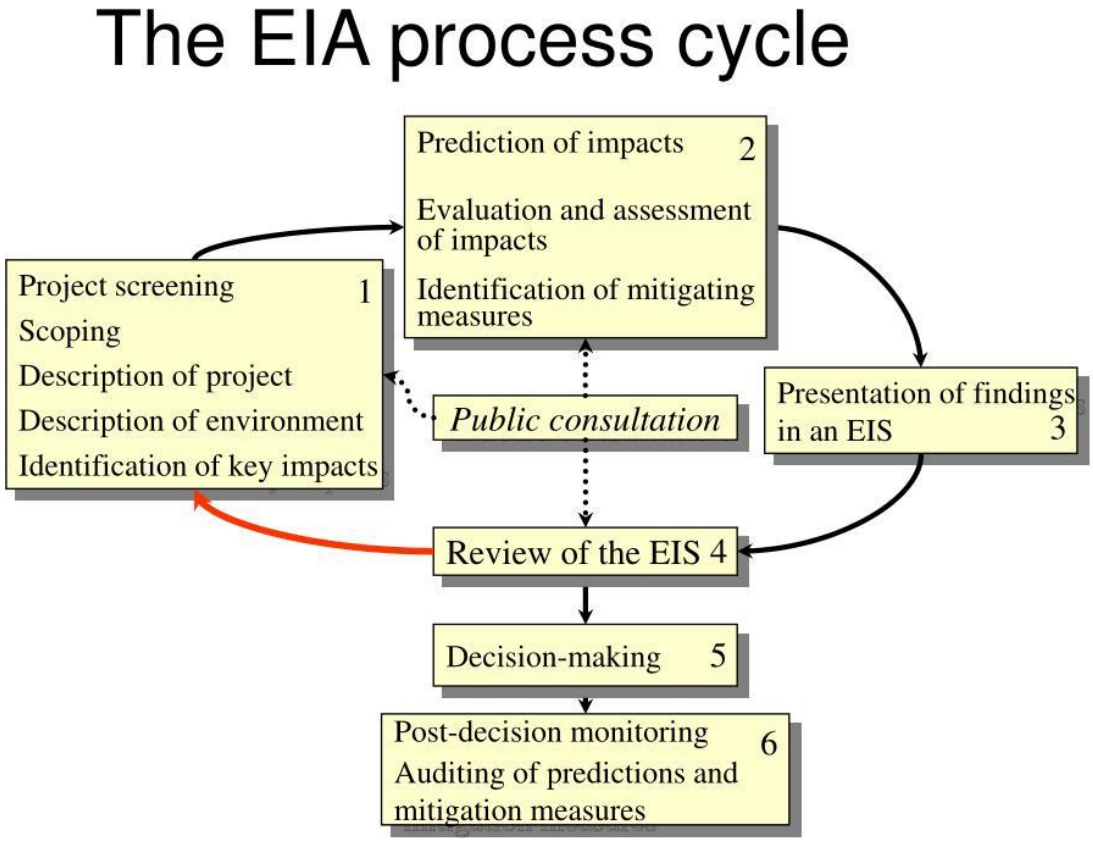
The correct order of steps in the Environmental Impact Assessment (EIA) process is:

1. **Analysis**: This involves gathering and analyzing data on the potential environmental impacts of the proposed project.
2. **Mitigation**: Identifying measures to avoid, reduce, or offset the adverse impacts of the project.
3. **EIA Reporting**: Preparing the Environmental Impact Statement (EIS) or EIA report, which documents the findings of the analysis and the proposed mitigation measures.

4. **Review**: The EIA report is reviewed by the relevant authorities and the public to ensure its adequacy and accuracy.
5. **Decision Making**: Based on the review, a decision is made on whether to approve, modify, or reject the project.

Therefore, the correct answer is:

3. A, D, C, E, B



Match the List-I with List-II

LIST-I ISO series		**LIST-II** Standards	
A.	ISO 14030	**I.**	Life Cycle Assessment
B.	ISO 14040	**II.**	Green Bonds
C.	ISO 14050	**III.**	Principles, requirements and guidelines
D.	ISO 14090	**IV.**	Vocabulary terms and definitions

1. A-II, B-I, C-IV, D-III
2. A-III, B-I, C-IV, D-II
3. A-II, B-IV, C-I, D-III
4. A-I, B-III, C-IV, D-II

ANSWER: A

ISO Series	Standard	Description
ISO 14030	Green Bonds	Standards related to green bonds, including requirements and guidelines for bonds used to finance environmental projects.
ISO 14040	Life Cycle Assessment	Standards providing the framework and requirements for conducting life cycle assessments, which evaluate the environmental impacts of products or services throughout their life cycle.
ISO 14050	Vocabulary Terms and Definitions	Standards providing clear and consistent definitions of terms used in environmental management.
ISO 14090	Principles, Requirements and Guidelines	Standards covering principles, requirements, and guidelines for climate change adaptation.

ISO Standard	Description
ISO 14001	Environmental management systems – Requirements with guidance for use
ISO 14004	Environmental management systems – General guidelines on implementation
ISO 14005	Environmental management systems – Guidelines for the phased implementation
ISO 14006	Environmental management systems – Guidelines for incorporating ecodesign
ISO 14015	Environmental management – Guidelines for environmental due diligence assessment
ISO 14020	Environmental labels and declarations – General principles
ISO 14031	Environmental management – Environmental performance evaluation – Guidelines
ISO 14045	Principles, requirements and guidelines for eco-efficiency assessment
ISO 14046	Environmental management – Water footprint – Principles, requirements and guidelines
ISO 14051	Environmental management – Material flow cost accounting – General framework
ISO 14064	Greenhouse gases – Quantification and reporting – Requirements and guidelines
ISO/TR 14073	Environmental management – Water footprint – Illustrative examples on how to apply ISO 14046
ISO 14084	Process diagrams for power plants
ISO/TS 14101	Surface characterization of gold nanoparticles for nanomaterial specific toxicity screening: FT-IR method
ISO/IEC 14102	Information technology – Guideline for the evaluation and selection of CASE tools
ISO/TR 14105	Document management – Change management for successful electronic document management system (EDMS) implementation
ISO 14117	Active implantable medical devices – Electromagnetic compatibility – EMC test protocols for implantable cardiac pacemakers, implantable cardioverter defibrillators and cardiac resynchronization devices
ISO 14132	Optics and photonics – Vocabulary for telescopic systems
ISO 14132-1	General terms and alphabetical indexes of terms in ISO 14132
ISO 14132-2	Terms for binoculars, monoculars and spotting scopes
ISO 14132-3	Terms for telescopic sights
ISO 14132-4	Terms for astronomical telescopes
ISO 14132-5	Terms for night vision devices
ISO/IEC 14136	Information technology – Telecommunications and information exchange between systems – Private Integrated Services Network – Specification, functional model and information flows – Identification supplementary services

People's Biodiversity Register (PBR) as per Biological Diversity Act 2002, is one of the main functions of

1. Biodiversity Management Committee (BMC)
2. State Biodiversity Board (SBB)
3. National Biodiversity Authority (NBA)
4. Village and Urban Local Bodies (V/ULB)

ANSWER: A

The People's Biodiversity Register (PBR), as per the Biological Diversity Act, 2002, is one of the main functions of the **Biodiversity Management Committee (BMC)**.

So, the correct answer is:

1. Biodiversity Management Committee (BMC)

Entity	Roles and Responsibilities
Biodiversity Management Committee (BMC)	- Prepare People's Biodiversity Register (PBR) containing comprehensive information on the availability and knowledge of local biological resources. - Promote conservation, sustainable use, and documentation of biodiversity. - Advise on matters relating to the identification of areas of biodiversity importance.
State Biodiversity Board (SBB)	- Regulate activities related to access to biological resources for commercial utilization. - Advise the state government on matters relating to the conservation and sustainable use of biodiversity. - Grant approval for requests for access to biological resources and associated knowledge.
National Biodiversity Authority (NBA)	- Implement provisions of the Biological Diversity Act at the national level. - Regulate access to biological resources and associated traditional knowledge. - Advise the central government on matters related to biodiversity conservation. - Grant approvals for access to biological resources by foreign individuals, companies, and institutions.
Village and Urban Local Bodies (V/ULB)	- Support the establishment of Biodiversity Management Committees (BMCs) at the local level. - Coordinate with BMCs to promote the conservation and sustainable use of biodiversity. - Facilitate the preparation and implementation of local biodiversity management plans.

Which of the following is the most recent contribution of India having the global impact?

1. Coalition for Disaster Resilient Infrastructure (CDRI)
2. International Big Cat Alliance (IBCA)
3. International Solar Alliance (ISA)
4. Mission LiFE

ANSWER: B

Which of the following categories of projects necessarily have to carry out EIA studies along with the 'Public Hearing' as per the MoEFCC, Government of India?

A. Category A

B. Category B

C. Category B1

D. Category B2

E. Category Al

Choose the most appropriate answer from the options given below:

1. A and B Only

2. A and C Only

3. A, B and D Only

4. C, D and E Only

ANSWER: B

The categories of projects that necessarily have to carry out Environmental Impact Assessment (EIA) studies along with the 'Public Hearing' as per the Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India, are:

- **Category A**
- **Category B1**

So, the correct answer is:

3. A and C

The concept of Nature Based Solution (NBS) is encouraged by-

1. International Geosphere Biosphere Programme (IGBP)
2. Pilot Analysis of Global Ecosystem (PAGE)
3. International Union for Conservation of Nature (IUCN)
4. International Human Dimensions Programme and Global Environmental Change (IHDP)

ANSWER: C

The concept of Nature-Based Solutions (NBS) is encouraged by the **International Union for Conservation of Nature (IUCN)**

Nature-Based Solutions (NBS)

Definition: Nature-Based Solutions (NBS) refer to the sustainable management and use of nature to tackle socio-environmental challenges such as climate change, water security, food security, human health, and disaster risk management. NBS aim to enhance and work with natural processes to improve resilience and mitigate the impacts of these challenges.

Encouragement by IUCN

The concept of Nature-Based Solutions is strongly encouraged by the **International Union for Conservation of Nature (IUCN)**. The IUCN is a global authority on the status of the natural world and the measures needed to safeguard it. They have been at the forefront of promoting NBS as a key approach to addressing global challenges.

Key Principles of NBS

1. **Sustainability**: NBS should promote sustainability by integrating ecological, social, and economic benefits.
2. **Biodiversity Conservation**: NBS should support biodiversity conservation and ecosystem restoration.

3. **Inclusivity**: NBS should involve and benefit local communities and stakeholders.
4. **Adaptability**: NBS should be adaptable to changing environmental and socio-economic conditions.

Examples of NBS

1. **Restoration of Wetlands**: Wetlands can act as natural water filters, reducing pollution and providing habitat for biodiversity. Restoring wetlands can also help mitigate flooding.
2. **Urban Green Spaces**: Planting trees and creating green spaces in urban areas can reduce heat island effects, improve air quality, and provide recreational spaces for communities.
3. **Mangrove Restoration**: Mangroves protect coastlines from erosion, provide habitat for marine life, and capture significant amounts of carbon dioxide.
4. **Agroforestry**: Integrating trees and shrubs into agricultural landscapes can improve soil health, increase biodiversity, and enhance agricultural productivity.

Organization	Role and Responsibility
International Geosphere-Biosphere Programme (IGBP)	- Conduct global and regional research on interactions between Earth's biological, chemical, and physical processes and human systems. - Enhance understanding of how these processes regulate the Earth system and their interactions. - Facilitate synthesis and capacity building in global change research.
Pilot Analysis of Global Ecosystems (PAGE)	- Analyze the condition of major ecosystems (freshwater, coastal, forest, grassland, and agroecosystems). - Develop indicators for assessing ecosystem services such as food, water, biodiversity, and carbon storage. - Highlight the need for sustainable management practices.
International Human Dimensions Programme on Global Environmental Change (IHDP)	- Study the human and societal aspects of global environmental change. - Integrate social science research into global change debates. - Address topics such as human security, urbanization, industrial transformation, and environmental governance.
International Union for Conservation of Nature (IUCN)	- Promote Nature-Based Solutions (NBS) to address global challenges like climate change, water security, food security, human health, and disaster risk management. - Develop frameworks and guidelines for implementing NBS. - Provide funding and support for NBS projects. - Raise awareness through advocacy and education. - Collaborate with governments, NGOs, and the private sector for NBS initiatives.

Which of the following is portal by MoEFCC, Government of India to automate the process of Environmental Clearance?

1. EIACP
2. ENVIS
3. e Green watch
4. PARIVESH

ANSWER: D

PARIVESH: Portal for Environmental Clearance

PARIVESH stands for **Proactive and Responsive facilitation by Interactive and Virtuous Environmental Single-window Hub**. It is a web-based, role-based workflow application developed by the Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India, in collaboration with the National Informatics Centre (NIC)2.

Key Features of PARIVESH

1. **Online Submission**: Proponents can submit proposals for Environment, Forest, Wildlife, and Coastal Regulation Zone (CRZ) clearances online.
2. **Monitoring and Tracking**: The portal allows for the online monitoring and tracking of proposals at each stage of the workflow.
3. **Role-Based Access**: Different roles are assigned to various stakeholders, including project proponents, state processing authorities, and central processing authorities.
4. **Status Updates**: Users can view the status of their proposals at each stage of the approval process.
5. **User Manuals and Forms**: The portal provides user manuals and updated forms to assist users in the submission process.

6. **Know Your Approval (KYA)**: A new functionality that allows users to check the status of their approvals.
7. **Helpdesk Support**: A toll-free helpdesk number is available for technical support and assistance.

Benefits of PARIVESH

- **Efficiency**: Automates the entire process of proposal submission and approval, reducing processing time and improving efficiency.
- **Transparency**: Provides real-time status updates and tracking, ensuring transparency in the approval process.
- **Accessibility**: Makes it easier for proponents to apply for clearances and for authorities to process and monitor proposals.
- **User-Friendly**: Designed to be user-friendly with comprehensive user manuals and support.

PARIVESH is a significant step towards streamlining the environmental clearance process in India, promoting transparency, and ensuring timely approvals for projects.

Which of the following states that "EIA as a national instrument shall be undertaken for the proposed activities that are likely to have significant adverse impact on the environment and are subject to a decision of a competent national authority"

1. Principle 17 of Rio Declaration (1992)
2. Article 14 of Convention on Biological Diversity (1993)
3. Article 10 of Environment (Protection) Act (1986)
4. Principle 18 of United State Environment Policy Act (1969)

ANSWER: A

The statement that "Environmental Impact Assessment (EIA) as a national instrument shall be undertaken for the proposed activities that are likely to have significant adverse impact on the environment and are subject to a decision of a competent national authority" is stated in **Principle 17 of the Rio Declaration (1992)**.

So, the correct answer is:

1. Principle 17 of Rio Declaration (1992)

Here's a table summarizing the **principles of the Rio Declaration** on Environment and Development, adopted in 1992:

Principle	Description
Principle 1	Human beings are at the center of concerns for sustainable development. They are entitled to a healthy and productive life in harmony with nature.
Principle 2	States have the sovereign right to exploit their own resources pursuant to their own environmental and developmental policies, and the responsibility to ensure that activities within their jurisdiction do not

Principle	Description
	cause damage to the environment of other States or areas beyond national jurisdiction.
Principle 3	The right to development must be fulfilled so as to equitably meet developmental and environmental needs of present and future generations.
Principle 4	In order to achieve sustainable development, environmental protection shall constitute an integral part of the development process and cannot be considered in isolation from it.
Principle 5	All states and all people shall cooperate in the essential task of eradicating poverty as an indispensable requirement for sustainable development, in order to decrease the disparities in standards of living and better meet the needs of the majority of the people of the world.
Principle 6	The special situation and needs of developing countries, particularly the least developed and those most environmentally vulnerable, shall be given special priority. International actions in the field of environment and development should also address the interests and needs of all countries.
Principle 7	States shall cooperate in a spirit of global partnership to conserve, protect and restore the health and integrity of the Earth's ecosystem. Developed countries acknowledge the responsibility that they bear in the international pursuit of sustainable development in view of the pressures their societies place on the global environment and of the technologies and financial resources they command.
Principle 8	To achieve sustainable development and a higher quality of life for all people, States should reduce and eliminate unsustainable patterns of production and consumption and promote appropriate demographic policies.
Principle 9	States should cooperate to strengthen endogenous capacity-building for sustainable development by improving scientific understanding through exchanges of scientific and technological knowledge, and by enhancing

Principle	Description
	the development, adaptation, diffusion, and transfer of technologies, including new and innovative technologies.
Principle 10	Environmental issues are best handled with the participation of all concerned citizens, at the relevant level.
Principle 11	States shall enact effective environmental legislation. Environmental standards, management objectives, and priorities should reflect the environmental and developmental context to which they apply.
Principle 12	States should cooperate to promote a supportive and open international economic system that would lead to economic growth and sustainable development in all countries, to better address the problems of environmental degradation.
Principle 13	States shall develop national law regarding liability and compensation for the victims of pollution and other environmental damage.
Principle 14	States should effectively cooperate to discourage or prevent the relocation and transfer to other States of any activities and substances that cause severe environmental degradation or are found to be harmful to human health.
Principle 15	In order to protect the environment, the precautionary approach shall be widely applied by States according to their capabilities.
Principle 16	National authorities should endeavor to promote the internalization of environmental costs and the use of economic instruments, taking into account the approach that the polluter should, in principle, bear the cost of pollution, with due regard to the public interest and without distorting international trade and investment.
Principle 17	Environmental Impact Assessment (EIA) as a national instrument shall be undertaken for proposed activities that are likely to have a significant adverse impact on the environment.
Principle 18	States shall immediately notify other States of any natural disasters or other emergencies that are likely to produce sudden harmful effects on

Principle	Description
	the environment of those States. Every effort shall be made by the international community to help States so afflicted.
Principle 19	States shall provide prior and timely notification and relevant information to potentially affected States on activities that may have a significant adverse transboundary environmental effect and shall consult with those States at an early stage and in good faith.
Principle 20	Women have a vital role in environmental management and development. Their full participation is therefore essential to achieve sustainable development.
Principle 21	The creativity, ideals, and courage of the youth of the world should be mobilized to forge a global partnership in order to achieve sustainable development and ensure a better future for all.
Principle 22	Indigenous people and their communities, and other local communities, have a vital role in environmental management and development because of their knowledge and traditional practices.
Principle 23	The environment and natural resources of people under oppression, domination, and occupation shall be protected.
Principle 24	Warfare is inherently destructive of sustainable development. States shall therefore respect international law providing protection for the environment in times of armed conflict and cooperate in its further development, as necessary.
Principle 25	Peace, development, and environmental protection are interdependent and indivisible.
Principle 26	States shall resolve all their environmental disputes peacefully and by appropriate means in accordance with the Charter of the United Nations.
Principle 27	States and people shall cooperate in good faith and in a spirit of partnership in the fulfillment of the principles embodied in this

Principle	Description
	Declaration and in the further development of international law in the field of sustainable development.

The Environment (Protection) Act, 1986 consists of 26 sections. Here's a brief summary of each section:

Section	Description
Section 1	Short title, extent, and commencement
Section 2	Definitions
Section 3	Power of Central Government to take measures to protect and improve the environment
Section 4	Appointment of officers and their powers and functions
Section 5	Power to give directions
Section 5A	Appeal to National Green Tribunal
Section 6	Rules to regulate environmental pollution
Section 7	Persons carrying on industry, operations, etc., not to allow emission or discharge of environmental pollutants in excess of the standards
Section 8	Persons handling hazardous substances to comply with procedural safeguards
Section 9	Powers of entry and inspection
Section 10	Sample and procedure to be followed in connection with the powers of entry and inspection
Section 11	Environmental laboratories

Section	Description
Section 12	Government Analysts
Section 13	Reports of Government Analysts
Section 14	Offences by companies
Section 14A	Penalty for contravention of section 7 or section 14
Section 14B	Penalty for contravention of sections 9, 10, and 11
Section 15	Cognizance of offences
Section 15A	Penalty for contravention by companies
Section 15B	Penalty for contravention by Government Departments
Section 15C	Adjudicating officer
Section 15D	Appeal
Section 16	Environmental Protection Fund
Section 16A	Accounts and audit of Fund
Section 17	Information, reports, or returns

Section	Description
Section 18	Authority constituted under Section 3 to be public authority under RTI Act
Section 19	Bar of jurisdiction
Section 20	Power to delegate
Section 21	Effect of other laws
Section 22	Power to make rules
Section 23	Rules made under this Act to be laid before Parliament

Which of the following is NOT linked to Biodiversity loss and ecosystem services?

1. Intergovernmental Science-Policy platform for Biodiversity and Ecosystem Services (IPBES)

2. Commission on Ecosystem Management (CEM)

3. Millennium Ecosystem Assessment (MEA)

4. The Economics of Ecosystem and Biodiversity (TEEB)

ANSWER: B

Option	Description
Intergovernmental Science-Policy Platform for Biodiversity and Ecosystem Services (IPBES)	IPBES assesses the state of biodiversity and ecosystem services, providing policymakers with scientific information to make informed decisions. It aims to bridge the gap between science and policy to promote conservation and sustainable use of biodiversity.
Commission on Ecosystem Management (CEM)	Part of the International Union for Conservation of Nature (IUCN), CEM focuses on promoting ecosystem-based approaches to the management of land, water, and living resources.
Millennium Ecosystem Assessment (MEA)	The MEA was a global initiative that assessed the consequences of ecosystem change for human well-being. It provided a scientific basis for actions needed to enhance the conservation and sustainable use of ecosystems and their services. The MEA highlighted the linkages between ecosystems and human well-being.
The Economics of Ecosystems and Biodiversity (TEEB)	TEEB is a global initiative focused on recognizing, demonstrating, and capturing the value of ecosystem services and biodiversity. It aims to mainstream the economic benefits of biodiversity into decision-making at all levels, promoting sustainable management of natural resources.

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Arrange the following conventions in correct chronological order :

(A) Ramsar

(B) CITES

(C) BONN

(D) BASEL

Choose the correct answer from the options given below :

(1) (A), (B), (C), (D)

(2) (B), (C), (D), (A)

(3) (C), (D), (B), (A)

(4) (D), (C), (B), (A)

ANSWER: A

To arrange the conventions in the correct chronological order, let's look at the years when each convention was established:

- **Ramsar Convention** (Convention on Wetlands) - 1971
- **CITES** (Convention on International Trade in Endangered Species of Wild Fauna and Flora) - 1973
- **BONN Convention** (Convention on the Conservation of Migratory Species of Wild Animals) - 1979
- **BASEL Convention** (Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal) - 1989

So, the correct answer is:

1. (A), (B), (C), (D)

Identify the correct statements about Strategic Environmental Assessment (SEA):

(A) This is a part of Environmental Impact Assessment (EIA).

(B) It is a systematic process for evaluating the environmental consequences of the proposed policy, plan and programme.

(C) It is useful in formulation of environmentally sound programme and for decision making.

(D) SEA should be undertaken by proponents of policy, plan and programme.

Choose the correct answer from the options given below :

(1) (A), (B), (E) Only

(2) (B), (C), (D) Only

(3) (A), (C), (D) Only

(4) (A), (B) Only

ANSWER: B

Let's evaluate each statement about Strategic Environmental Assessment (SEA):

(A) This is a part of Environmental Impact Assessment (EIA).

- **Incorrect**: SEA and EIA are related but separate processes. SEA is applied to policies, plans, and programs, while EIA is project-specific.

(B) It is a systematic process for evaluating the environmental consequences of the proposed policy, plan and programme.

- **Correct**: SEA is indeed a systematic process for evaluating the environmental consequences of policies, plans, and programs.

(C) It is useful in formulation of environmentally sound programme and for decision making.

- **Correct**: SEA helps in formulating environmentally sound programs and aids in decision-making.

(D) SEA should be undertaken by proponents of policy, plan and programme.

- **Correct**: SEA should be conducted by those who propose the policy, plan, or program to ensure environmental considerations are integrated from the beginning.

Given this evaluation, the correct answer is:

2. (B), (C), (D) Only

Match List - I with List - II.

LIST-I Protocol		**LIST-II** Features	
A.	Nagoya	**I.**	Procedures for Living Modified Organisms intended to be introduced into the environment
B.	Cartagena	**II.**	Access and Benefit Sharing of genetic resources
C.	Kyoto	**III.**	Reduction in the production of the CFCs
D.	Montreal	**IV.**	Control of Green house gases emissions

Choose the correct answer from the options given below:

(1) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

(2) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

(3) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(4) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

ANSWER: C

Let's match the protocols with their respective features:

- **Nagoya Protocol**: Access and Benefit Sharing of genetic resources.
- **Cartagena Protocol**: Procedures for Living Modified Organisms intended to be introduced into the environment.
- **Kyoto Protocol**: Control of Greenhouse gases emissions.
- **Montreal Protocol**: Reduction in the production of CFCs (Chlorofluorocarbons).

Therefore, the correct answer is:

3. (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

Under the Wild Life Protection Act 1972, Vermin are wild animals specified in:

(1) Schedule I

(2) Schedule II

(3) Schedule IV

(4) Schedule V

ANSWER: D

Under the Wild Life Protection Act, 1972, vermin are wild animals specified in **Schedule V**

Key Points

66 Sections

7 Chapters

6 Schedules

WPA passed by Article 272

Boards under WPA

National Board for Wildlife – 2003

National Tiger Conservation Authority – 2006

Central Zoo Authority - 1993

WILDLIFE PROTECTION ACT, 1972

- The **Wild Life (Protection) Act, of 1972** provides a **legal framework for the protection of various species of wild animals and plants**, management of their habitats, regulation, and control of trade in wild animals, plants, and products made from them
- The act also lists schedules of plants and animals that are afforded varying degrees of protection and monitoring by the government.
- India's entry to the **CITES (Convention on International Trade in Endangered Species of Wild Fauna and Flora)** was made easier by the Wildlife Act.
- Earlier, **Jammu and Kashmir was not covered by the Wildlife Protection Act of 1972.** The Indian Wildlife Protection Act now applies to J&K as a result of the **reorganization act.**

- **Constitutional Provisions for the Wildlife Act:**
  - The **42nd Amendment Act,** 1976, **Forests and Protection of Wild Animals and Birds was transferred from State to Concurrent List.**
  - Article 51 A (g) of the Constitution states that it shall be the **fundamental duty** of **every citizen to protect and improve the natural environment including forests and Wildlife.**
  - Article 48 A in the **Directive Principles of State policy, mandates that the State shall endeavor to protect and improve the environment and to safeguard the forests and wildlife of the country.**
- **Schedules under the Act:**
  - **Schedule I:**
    - It covers **endangered species that need rigorous protection.**
    - A person is **liable to the harshest penalties for violation** of the law under this Schedule.
    - Species under this Schedule **are prohibited to be hunted throughout India,** except under threat to human life or in case of a disease that is beyond recovery.
    - Some of the animals listed under Schedule I include **the Black Buck, Snow Leopard, Himalayan Bear and Asiatic Cheetah.**
  - **Schedule II:**
    - **Animals under this list are also accorded high protection with the prohibition on their trade**.
    - Some of the animals listed under Schedule II include **Assamese Macaque, Himalayan Black Bear and Indian Cobra.**
  - **Schedule III & IV:**
    - Species that are **not endangered are included under Schedule III and IV.**
    - This includes **protected species with hunting prohibited but the penalty for any violation is less compared to the first two schedules.**
    - Animals protected under Schedule III include Chital (spotted deer), Bharal (blue sheep), Hyena, and Sambhar (deer).

- Animals protected under Schedule IV include **Flamingo, Hares, Falcons, Kingfishers, Magpie, and Horseshoes Crabs.**

- **Schedule V:**

- This schedule contains animals that are considered as **vermin** (small wild animals that carry disease and destroy plants and food). These animals can be hunted.

- It includes only four species of wild animals: **Common Crows, Fruit Bats, Rats, and Mice.**

- **Schedule VI:**

- It provides for **regulation in the cultivation of a specified plant** and restricts its possession, sale, and transportation.

- Both cultivation and trade of specified plants can only be carried out with the prior permission of the competent authority.

- Plants protected under Schedule VI include **Beddomes' cycad (Native to India), Blue Vanda (Blue Orchid), Red Vanda (Red Orchid), Kuth (Saussurea lappa), Slipper orchids (Paphiopedilum spp.) and Pitcher plant (Nepenthes khasiana).**

- **Bodies Constituted under the Act:**

- **National Board for Wildlife (NBWL):**

- **NBWL** serves as an **apex body for the review of all wildlife-related matters** and for the approval of projects in and around national parks and sanctuaries.

- **State Board for Wildlife (SBWL):**

- The Chief Minister of the state/UT is the chairperson of the board.

- **Central Zoo Authority:**

- The **Central Zoo Authority** consists of a total 10 members including the Chairperson and a Member-Secretary.

- The **authority provides recognition to zoos and is also tasked with regulating the zoos across the country.**

- It lays down **guidelines and prescribes rules under which animals may be transferred among zoos nationally and internationally.**

- **National Tiger Conservation Authority (NTCA):**

- Following the recommendations of the Tiger Task Force, **NTCA** was constituted in 2005 for strengthening tiger conservation.
- The Union Environment Minister is the Chairperson of NTCA and the State Environment Minister is the Vice-Chairperson.
- The Central Government on the recommendations of NTCA declares an area as a **Tiger Reserve.**

- **Wildlife Crime Control Bureau (WCCB):**

- The act provided for the constitution of **WCCB** to combat organized wildlife crime in the country.

- **Protected Areas Under the Act:**

- There are five types of protected areas under the Act which are: **Sanctuary, National Parks, Conservation Reserves, Community Reserves and Tiger Reserves.**

Important Amendments done to the Act:

-

Wildlife (Protection) Amendment Act of 1991:

- This amendment **strengthened the penalties and fines for wildlife-related offences** and also **introduced provisions for the protection of endangered species.**

Wildlife (Protection) Amendment Act of 2002:

- This amendment introduced the **concept of community reserves and conservation reserves as protected areas.**

Wildlife (Protection) Amendment Act of 2006:

- This amendment dealt with the issue of **human-wildlife conflict** and provided for the creation of a **National Tiger Conservation Authority (NTCA)** to manage and **protect tiger reserves.**

- It also made provisions for the creation of a **Tiger and Other Endangered Species Crime Control Bureau** to deal with wildlife-related crimes.

- **Wildlife (Protection) Amendment Act, 2022:**

- The Act seeks to **increase the species** protected under the law and implement CITES.
- The **number of schedules has been reduced to four:**
- **Schedule I containing animal species enjoying the highest level of protection.**
- **Schedule II for animal species subject to a lesser degree of protection.**
- **Schedule III for protected plant species,** and
- **Schedule IV for scheduled specimens under CITES.**
- The Act permits the **use of elephants** for **'religious or any other purposes'.**
- The penalties have also been increased for general and specially protected animals' violations.

JUNE 2024 Cancelled

The recently enacted Wild Life (Protection) Amendment Act, 2022 aligns itself with which of the following conventions?

(1) CBD: Convention on Biological Diversity

(2) CITES: Convention on International Trade in Endangered Species of Wild Fauna and Flora

(3) CMS: Convention on the Conservation of Migratory Species of Wild Animals

(4) TRAFFIC: Trade Records Analysis of Flora and Fauna in Commerce

ANSWER: B

Recently, the **Rajya Sabha** passed the **Wild Life (Protection) Amendment Bill, 2022** which seeks to give effect to India's obligations under the **Convention on International Trade on Endangered Species of Wild Fauna and Flora ('CITES').**

What is the Objective of Bill?

- **Protection of Endangered Species:** Bill seeks to **enhance punishment for illegal Wildlife trade .**
- **Better Management of Protected Areas:** It provides for certain permitted activities like **grazing or movement of livestock** and **Bonafide use of drinking and household water** by **local communities.**
- **Protection of Forest Lands**: It is so critical because it equally inculcates in itself the **protection of rights of the people** who have been residing there since ages.

Which of the following are listed as main objective of Convention on Biological Diversity (CBD), 1993?

A. Conservation of biological diversity

B. Establishing more predictable conditions for access to genetic resources

C. Fair and equitable sharing of the benefits arising out of the utilization of genetic resources

D. Protect biological diversity from the potential risks posed by living modified organisms resulting from modern biotechnology

E. Sustainable use of the components of biological diversity

Choose the most appropriate answer from the options given below:

(1) A, B, C only

(2) A, C, E only

(3) B, C, E only

(4) C, D, E only

ANSWER: B

The main objectives of the Convention on Biological Diversity (CBD), 1993 are:
A. Conservation of biological diversity E. Sustainable use of the components of biological diversity C. Fair and equitable sharing of the benefits arising out of the utilization of genetic resources
Options B and D are not listed as main objectives of the CBD

Which of the following is not part of National Biological Diversity Act 2002?

(1) Biodiversity Management Committee (BMC)

(2) National Biodiversity Authority (NBA)

(3) Peoples Biodiversity Register (PBR)

(4) Traditional Knowledge Digital Library (TKDL)

ANSWER: D

The **Traditional Knowledge Digital Library (TKDL)** is not part of the National Biological Diversity Act 2002. The other three components—**Biodiversity Management Committee (BMC)**, **National Biodiversity Authority (NBA)**, and **Peoples Biodiversity Register (PBR)**—are integral to the Act.

Which of the following is a portal maintained by the Ministry of Environment Forest and Climat Change (MoEFCC) to automate the process of Environmental Clearance (EC) bringing down the overall time taken in grant of clearance as per the provision of the Environmental Impact Assessment (EIA) ?

(1) ANTHRA

(2) CHINTAN

(3) DISHA

(4) PARIVESH

ANSWER: D

The portal maintained by the Ministry of Environment, Forest and Climate Change (MoEFCC) for automating the process of Environmental Clearance (EC) is called **PARIVESH**. It streamlines the entire EC process, from application submission to clearance grant.

Match List - I with List - II.

List - I	List - II
A. ISO 14010	I. Environmental Performance Evaluation
B. ISO 14020	II. Life Cycle Assessment
C. ISO 14030	III. Environmental Labelling
D. ISO 14040	IV. Environmental Auditing

Choose the correct answer from the options given below:

(1) A-IV, B-III, C-I, D-II

(2) A-II, B-1, C-1, D-IV

(3) A-IV, B-1, C-III, D-II

(4) A-I, B-IV, C-II, D-III

ANSWER: A

Let's match the ISO standards with their corresponding descriptions:

1. **ISO 14010**: Environmental Performance Evaluation
2. **ISO 14020**: Principles and general requirements for environmental statements and programmes for products
3. **ISO 14030**: Environmental Labelling
4. **ISO 14040**: Life Cycle Assessment

So, the correct answer is **(1) a-iv, b-iii, c-i, d-ii**

Which of the following step refers to the process to determine detailed and comprehensive Term of Reference (ToR) addressing all relevant environmental concerns for the preparation of a Environmental Impact Assessment (EIA) Report in respect of project and activity for whic Environmental Clearance (EC) is sought?

(1) Appraisal

(2) Public Consultation

(3) Scoping

(4) Screening

ANSWER: C

The step that refers to the process of determining detailed and comprehensive Terms of Reference (ToR) addressing all relevant environmental concerns for the preparation of an Environmental Impact Assessment (EIA) Report is **Scoping**.

So, the correct answer is:

3. Scoping

Arrange the following policies related to environmental in chronological order (from the first to the latest)

A. National Agriculture Policy

B. National Clean Air Programme

C. National Environmental Policy

D. National Forest Policy

E. National Water Policy

Choose the correct answer from the options given below:

(1) E, D, A, B, C

(2) D, A, E, C, B

(3) D, E, A, C, B

(4) A, E, D, B. C

ANSWER: B

Let's arrange these environmental policies in chronological order:

1. **National Forest Policy (D)**: The National Forest Policy was formulated in 1988, emphasizing sustainable forest management and conservation.
2. **National Agriculture Policy (A)**: The National Agriculture Policy was introduced in 2000, focusing on agricultural development and resource utilization.
3. **National Water Policy (E)**: The National Water Policy was established in 2002, addressing water resource management and conservation.
4. **National Environmental Policy** ©: The National Environmental Policy, enacted in 2006, outlines principles for environmental protection and sustainable development.
5. **National Clean Air Programme (B)**: The National Clean Air Programme, launched in January 2019, aims to reduce air pollution by 20-30% by 2024.

Therefore, the correct order is **D, A, E, C, B** (option 2).

Match List - I with List - II.

List-1 Convention	List – II Environmental Issue
A. Basel Convention	I. Persistent Organic Pollutants (POPs)
B. Bonn Convention	II. Protection of Ozone Layer
C. Stockholm Convention	III. Conservation of Migratory Species of Wild Animals
D. Vienna Convention	IV. Control of transboundary movements of Hazardous Wastes and their Disposal

Choose the correct answer from the options given below:

(1) A-IV, B-I, C-III, D-II

(2) A-IV, B-II, C-III, D-1

(3) A-IV, B-III, C-I, D-II

(4) A-II, B-1, C-III, D-IV

ANSWER: C

Let's match the conventions with their corresponding environmental issues:

- **Basel Convention**: Control of transboundary movements of hazardous wastes and their disposal.
- **Bonn Convention**: Conservation of migratory species of wild animals.
- **Stockholm Convention**: Persistent Organic Pollutants (POPs).
- **Vienna Convention**: Protection of ozone layer.

Therefore, the correct answer is:

3. (A-IV, B-III, C-I, D-II)

Which of the following countries hosted the United Nations Climate Change Conference referred as Conference of the Parties (CoP), held in the framework of the United Nations Framework Convention on Climate Change (UNFCCC) for the maximum times?

(1) Denmark

(2) Germany

(3) South Africa

(4) Sweden

ANSWER: B

List of conferences and locations[edit]

No.	Year	Name	Alternative name	Location	Country	Regional Groups[7]
1	1995	COP 1		Berlin	Germany	Western Europe and Other Group (WEOG)
2	1996	COP 2		Geneva	Switzerland	WEOG
3	1997	COP 3		Kyoto	Japan	Asia and Pacific Group
4	1998	COP 4		Buenos Aires	Argentina	Group of Latin America and the Caribbean (GRULAC)
5	1999	COP 5		Bonn	Germany	WEOG

6	2000	COP 6		The Hague	Netherlands	WEOG
7	2001	COP 6-2		Bonn	Germany	WEOG
8	2001	COP 7		Marrakech	Morocco	Africa Group
9	2002	COP 8		New Delhi	India	Asia and Pacific Group
10	2003	COP 9		Milan	Italy	WEOG (COP President: Hungary, Eastern Europe Group)
11	2004	COP 10		Buenos Aires	Argentina	GRULAC
12	2005	COP 11	CMP 1	Montreal	Canada	WEOG
13	2006	COP 12	CMP 2	Nairobi	Kenya	Africa Group
14	2007	COP 13	CMP 3	Bali	Indonesia	Asia and Pacific Group
15	2008	COP 14	CMP 4	Poznań	Poland	Eastern Europe Group
16	2009	COP 15	CMP 5	Copenhagen	Denmark	WEOG
17	2010	COP 16	CMP 6	Cancún	Mexico	GRULAC

18	2011	COP 17	CMP 7	Durban	South Africa	Africa Group
19	2012	COP 18	CMP 8	Doha	Qatar	Asia and Pacific Group
20	2013	COP 19	CMP 9	Warsaw	Poland	Eastern Europe Group
21	2014	COP 20	CMP 10	Lima	Peru	GRULAC
22	2015	COP 21	CMP 11	Paris	France	WEOG
23	2016	COP 22	CMP 12 / CMA 1	Marrakech	Morocco	Africa Group
24	2017	COP 23	CMP 13 / CMA 1-2	Bonn	Germany	WEOG (COP President: Fiji, Asia and Pacific Group)
25	2018	COP 24	CMP 14 / CMA 1-3	Katowice	Poland	Eastern Europe Group
26	2019	SB50		Bonn	Germany	WEOG
27	2019	COP 25	CMP 15 / CMA 2	Madrid	Spain	WEOG (COP President: Chile, GRULAC)
28	2021	COP 26	CMP 16 / CMA 3	Glasgow	United Kingdom	WEOG

29	2022	COP 27	CMP 17 / CMA 4	Sharm El Sheikh	Egypt	Africa Group
30	2023	COP 28	CMP 18 / CMA 5	Dubai	United Arab Emirates	Asia and Pacific Group
31	2024	COP 29	CMP 19 / CMA 6	Baku	Azerbaijan	Eastern Europe Group[8]
32	2025	COP 30	CMP 20 / CMA 7	Belém	Brazil	GRULAC
33	2026	COP 31	CMP 21 / CMA 8	TBC	TBC	WEOG Australia and Pacific Nations

December 2023

Which of the following are project/product based tools in Environmental Management ?

(A) Cleaner Production

(B) Design for Environment

(C) Life Cycle Assessment

(D) Risk Assessment

(E) Technology Assessment

Choose the correct answer from the options given below :

(1) (A), (B), (C) Only

(2) (B), (C), (D) Only

(3) (C), (D), (E) Only

(4) (A), (C), (E) Only

ANSWER: B

As per the Biological Diversity (Amendment) Act 2023, the "Farmer's variety" means a variety which:

(A) has originated and persisted under cultivation or was specifically used for the purpose of cultivation

(B) was developed, grown and exchanged informally among farmers.

(C) has been traditionally cultivated and evolved by the farmers in their field

(D) is a wild relative or landrace of a variety about which the farmers possess the common knowledge

(E) was grown by ancient farmers and their successors

Choose the correct answer from the options given below :

(1) (A), (B) Only

(2) (B), (C) Only

(3) (C), (D) Only

(4) (D), (E) Only

ANSWER: C

Leopold interaction-matrix used in Environmental Impact Assessment (EIA) includes:

(1) Adaptation and Mitigation

(2) Benefits and Detriments

(3) Losses and Damages

(4) Magnitude and Importance

ANSWER: D

The Leopold interaction-matrix used in Environmental Impact Assessment (EIA) includes Magnitude and Importance.

So, the correct answer is:

4. **Magnitude and Importance**

Leopold Interaction-Matrix in Environmental Impact Assessment (EIA)

The **Leopold Matrix** is a tool developed by Luna Leopold and colleagues in 1971 to evaluate the environmental impact of various projects. It is widely used in Environmental Impact Assessment (EIA) due to its systematic and comprehensive approach.

Structure of the Leopold Matrix

The Leopold Matrix is a two-dimensional grid that lists project actions along one axis and environmental factors along the other. Each cell in the matrix represents a potential interaction between a specific project action and an environmental factor.

The matrix typically includes:

- **100 project actions**: These actions can range from land clearing, construction, and operation of facilities to the disposal of wastes.

- **88 environmental factors**: These factors encompass a wide range of biophysical, ecological, social, and economic elements that might be affected by the project.

Key Components

1. Project Actions:

- Actions that involve direct physical changes to the environment (e.g., excavation, construction).
- Operational actions that may have indirect environmental impacts (e.g., emissions, waste discharge).

2. Environmental Factors:

- Physical and chemical aspects (e.g., air quality, water quality, soil stability).
- Biological aspects (e.g., flora, fauna, habitat).
- Social and economic aspects (e.g., public health, cultural heritage, economy).

Evaluation Criteria

Magnitude: This represents the scale or extent of the impact, ranging from minor to major. **Importance**: This represents the significance of the impact, considering factors such as the sensitivity of the affected environment and the value placed on it by society.

Each cell in the matrix is assigned two values:

1. **Magnitude of impact**: On a scale from 1 (least impact) to 10 (most impact).
2. **Importance of impact**: Also on a scale from 1 to 10.

By filling in these values for each interaction, the matrix provides a clear and detailed assessment of the environmental impacts associated with the project.

Example of a Leopold Matrix

Here is a simplified example to illustrate the concept:

Environmental Factors	Land Clearing	Construction	Operation	Waste Disposal
Air Quality	7, 6	8, 7	6, 5	5, 4
Water Quality	6, 5	7, 6	5, 4	8, 7
Soil Stability	9, 8	8, 7	4, 3	6, 5
Flora and Fauna	8, 7	9, 8	7, 6	5, 4
Public Health	5, 4	6, 5	8, 7	7, 6

Advantages of the Leopold Matrix

- **Systematic Approach**: Provides a structured method to identify and evaluate environmental impacts.
- **Comprehensive**: Covers a wide range of project actions and environmental factors.
- **Visual Representation**: Offers a clear visual representation of the interactions and their significance.

Limitations of the Leopold Matrix

- **Complexity**: Can be complex and time-consuming to complete for large projects with many actions and factors.
- **Subjectivity**: The assignment of magnitude and importance values can be subjective and may vary among evaluators.

Which of the following Summit/Convention is known to be the first to aim at 20 percent cut in the emission of CO2 by 2005 AD but the developed countries backed out from the agreement on the pretext of non-availability of reliable data of emission of GHGs?

(1) World Climate Summit (Geneva), 1979

(2) Rio Summit (Brazil), 1992

(3) Toronto Summit (Canada), 1988

(4) Vienna Convention (Austria), 1985

ANSWER: C

The correct answer is:

3. Toronto Summit (Canada), 1988

The Toronto Summit in 1988 was the first international meeting to aim for a 20 percent cut in CO2 emissions by 2005. However, developed countries backed out of the agreement, citing the non-availability of reliable data on greenhouse gas emissions

Summit/Convention	Description
World Climate Summit (Geneva), 1979	The first World Climate Conference, held in Geneva in 1979, was organized by the World Meteorological Organization (WMO). It focused on understanding climate variability and change and laid the groundwork for future international climate initiatives.
Rio Summit (Brazil), 1992	Also known as the Earth Summit, it was held in Rio de Janeiro, Brazil, in 1992. The summit led to the adoption of key environmental agreements, including the United Nations Framework Convention on Climate Change (UNFCCC), the Convention on Biological Diversity (CBD), and Agenda 21.
Toronto Summit (Canada), 1988	Known as the "Toronto Conference on the Changing Atmosphere: Implications for Global Security," it was held in 1988. The summit aimed for a 20 percent cut in CO2 emissions by 2005, but developed countries backed out due to the lack of reliable data on greenhouse gas emissions.
Vienna Convention (Austria), 1985	The Vienna Convention for the Protection of the Ozone Layer was adopted in 1985. It provided a framework for international cooperation to protect the ozone layer and led to the development of the Montreal Protocol, which aimed to reduce the production of substances that deplete the ozone layer.

Arrange the following scoping methods used in EIA, in the order of advancement:

(A) Analogs

(B) Matrices

(C) Checklists

(D) Overlays

(E) Networks

Choose the correct answer from the options given below :

(1) (C), (E), (B), (A), (D)

(2) (C), (B), (E), (D), (A)

(3) (B), (C), (E), (A), (D)

(4) (C), (D), (E), (A), (B)

ANSWER: B

To arrange the scoping methods used in Environmental Impact Assessment (EIA) in the order of advancement, we need to consider the evolution and complexity of these methods:

- **Checklists**: Simple, early method to identify potential impacts.
- **Matrices**: More advanced than checklists, showing interactions between project activities and environmental components.
- **Networks**: Further advancement, showing more complex relationships and pathways of impacts.
- **Overlays**: Use of geographical data layers to identify potential impacts visually.
- **Analogs**: Comparatively newer method, using similar projects as references to predict impacts.

Therefore, the correct answer is:

2. (C), (B), (E), (D), (A)

Which of the following are part of life cycle assessment (LCA) of a product?

(A) Raw materials

(B) Product manufacturing

(C) Packaging and distribution

(D) Disposal of the product

(E) Product use

Choose the correct answer from the options given below:

(1) (B), (C) and (D) Only

(2) (A), (B), (C) and (D) Only

(3) (B), (C), (D) and (E) Only

(4) (A), (B), (C), (D) and (E)

ANSWER: D

Life Cycle Assessment (LCA) is a systematic process to evaluate the environmental impacts of a product throughout its entire life cycle, from raw materials to disposal. The stages included in an LCA are:

(A) Raw materials (B) Product manufacturing (C) Packaging and distribution (D) Disposal of the product (E) Product use

Therefore, the correct answer is:

4. (A), (B), (C), (D), and (E)

Which of the following Articles of the Constitution of India refers to protection and improvement of environment and safeguarding of forests and wildlife?

(1) 39 A

(2) 43 A

(3) 48 A

(4) 51 A

ANSWER: C

The Articles of the Constitution of India that refer to the protection and improvement of the environment and safeguarding of forests and wildlife are **Article 48A** and **Article 51A(g)**.

- **Article 48A**: "The State shall endeavour to protect and improve the environment and to safeguard the forests and wildlife of the country."
- **Article 51A(g)**: "It shall be the duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers, and wildlife, and to have compassion for living creatures."

Given the options provided, the correct answer is:

3. Article 48A

Which of the following is not used for elimination of the non-feasible set of alternative in multi attribute decision making ?

(1) Compromise programming

(2) Conjunctive Screening

(3) Compensatory screening

(4) Lexicographic screening

ANSWER: A

In multi-attribute decision-making, different techniques are used for evaluating and eliminating alternatives based on various attributes. Among the options provided, **Compromise Programming** is not primarily used for the elimination of non-feasible alternatives. Instead, it is a method used to find a solution that represents the best compromise among multiple conflicting criteria.

So, the correct answer is:

1. Compromise programming

Option	Description
Compromise Programming	A method used to find a solution that represents the best compromise among multiple conflicting criteria. It focuses on minimizing the maximum deviation from the ideal solution, rather than eliminating non-feasible alternatives.
Conjunctive Screening	An elimination method where alternatives are screened based on whether they meet a set of minimum threshold criteria for all attributes. Alternatives failing to meet any one of the criteria are eliminated.
Compensatory Screening	A method where poor performance in one attribute can be compensated by good performance in another attribute. Alternatives are evaluated based on a weighted sum of attributes, allowing trade-offs between attributes.
Lexicographic Screening	An elimination technique where attributes are ranked in order of importance. Alternatives are compared based on the most important attribute first, and ties are resolved by moving to the next most important attribute. This process continues until a decision is made.

Given below are two statements:

Statement (I): Strategic Environmental Assessment (SEA) helps to build a desirable future, it is not an attempt to know the future.

Statement (II): Environmental Impact Assessment (EIA) is performed through the preparation and development of Policies, Plans, Programs and projects.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

ANSWER: C

Let's analyze both statements:

Statement (I): Strategic Environmental Assessment (SEA) helps to build a desirable future, it is not an attempt to know the future.

- **Correct**: SEA is a proactive process that aims to integrate environmental considerations into policies, plans, and programs to promote sustainable development, focusing on creating a desirable future rather than predicting it.

Statement (II): Environmental Impact Assessment (EIA) is performed through the preparation and development of Policies, Plans, Programs, and projects.

- **Incorrect**: EIA is project-specific and focuses on assessing the environmental impacts of individual projects rather than policies, plans, and programs. SEA is the process used for the broader scope of policies, plans, and programs.

Given this analysis, the most appropriate answer is:

3. Statement I is correct but Statement II is incorrect

In April 2016, the UN General Assembly has proclaimed the UN decade of to 2025.

(1) action on nutrition

(2) millets

(3) plastic free world

(4) fisheries and aquaculture

ANSWER: A

In April 2016, the UN General Assembly proclaimed the UN Decade of Action on Nutrition to 2025.

So, the correct answer is:

1. action on nutrition

Here's a table listing some of the decades proclaimed by the United Nations:

Decade	Focus Area	Years
UN Decade of Action on Nutrition	Action on nutrition	2016–2025
United Nations Decade of Ecosystem Restoration	Ecosystem restoration	2021–2030
United Nations Decade of Action on Biodiversity	Biodiversity	2021–2030
United Nations Decade of Action on Desertification	Desertification	2010–2020
United Nations Decade of Action on Biodiversity	Biodiversity	2021–2030

Decade	Focus Area	Years
United Nations Decade of Action on Sustainable Energy for All	Sustainable energy for all	2014–2024
United Nations Decade of Action on Desertification	Desertification	2010–2020

Environmental Scientists and resource managers should adopt which of the following study (design) for documenting present state of the environment to establish future changes resulting from unforeseen events?

(1) Baseline studies

(2) Irregular Monitoring

(3) Regular Monitoring

(4) Targeted studies

ANSWER: A

To document the present state of the environment and establish future changes resulting from unforeseen events, **Baseline studies (1)** is generally the most effective approach

1. **Baseline Studies**: These are essential for establishing the current state of the environment. They provide a comprehensive snapshot of conditions at a specific point in time, which serves as a reference for future comparisons.

Irregular Monitoring (2) might miss critical changes due to its sporadic nature, and **Targeted Studies (4)** focus on specific aspects rather than the overall state, which might not provide a complete picture.

JUNE 2023

Life cycle assessment of a product includes:

A. Raw material extraction

B. Product manufacturing

C. Product remanufacture

D. Packaging and distribution

E. Product disposal

Choose the most appropriate answer from the options given below

1. B and D only

2. B, D and E only

3. A, B and E only

4. A, B, C, D and E only

ANSWER: D

The most appropriate answer for a comprehensive life cycle assessment (LCA) of a product, which includes all the stages from raw material extraction to product disposal, is:

4. A, B, C, D and E only

A thorough LCA considers:

- **Raw material extraction (A)**: Sourcing the raw materials needed for the product.
- **Product manufacturing (B)**: The process of converting raw materials into the final product.

- **Product remanufacture (C)**: Reusing, refurbishing, or remanufacturing parts or products to extend their lifecycle.
- **Packaging and distribution (D)**: The packaging of the product and its transportation to the market.
- **Product disposal (E)**: The end-of-life stage, which includes disposal, recycling, or waste management.

By covering all these stages, an LCA can provide a complete picture of the environmental impact of a product throughout its life.

Match List I with List II

LIST-I		LIST-II	
A.	Disaster Management Act	**I.**	1987
B.	Kyoto Protocol	**II.**	2008
C.	Montreal Protocol	**III.**	2005
D.	National Action Plan on climate change	**IV.**	1997

Choose the correct answer from the options given below:

1. A-I, B-II, C-III, D-IV

2. A-II. B-III, C-IV. D-I

3. A-III, B-IV, C-I, D-II

4. A-IV, B-III, C-II, D-I

ANSWER: C

Let's match the items from List-I with the correct years from List-II:

A. Disaster Management Act - 2005 (III) B. Kyoto Protocol - 1997 (IV) C. Montreal Protocol - 1987 (I) D. National Action Plan on climate change - 2008 (II)

So, the correct answer is:

3. A-III, B-IV, C-I, D-II

Match List I with List II

LIST-I (Acts)		**LIST-II**	
A.	Biological Diversity Act	**I.**	1972
B.	Environmental (Protection) Act	**II.**	1982
C.	Indian forest Act (Revised)	**III.**	1986
D.	Wildlife Protection Act	**IV.**	2002

Choose the correct answer from the options given below: C

1. A-IV, B-III, C-I, D-II
2. A-IV, B-II. C-I, D-III
3. A-IV, B-III, C-II, D-I
4. A-III, B-IV, C-II, D-I

ANSWER: C

Let us match the items from List-I with the correct years from List-II:

A. Biological Diversity Act - **2002** (IV)

B. Environmental (Protection) Act - **1986** (III)

C. Indian Forest Act (Revised) - **1982** (II)

D. Wildlife Protection Act - **1972** (I)

So, the correct answer is:

3. A-IV, B-III, C-II, D-I

"The formalized, systematic and comprehensive process of evaluating the environmental impacts of a policy, programme, plan or its alternatives as well as using the findings in publicly accountable decision making is called:

1. Environmental Impact Assessment (EIA)
2. Integrated Impact Assessment (IIA)
3. Ecological Impact Assessment (ECIA)
4. Strategic Environmental Impact Assessment (SEIA)

ANSWER: D

The correct answer is:

4. Strategic Environmental Impact Assessment (SEIA)

Strategic Environmental Impact Assessment (SEIA) is the formalized, systematic, and comprehensive process of evaluating the environmental impacts of a policy, programme, or plan, as well as its alternatives. It also involves using the findings in a way that is publicly accountable in decision-making processes.

Term	Explanation
Environmental Impact Assessment (EIA)	A formal process of evaluating the potential environmental effects of a proposed project or development. It is used to ensure that decision-makers consider environmental impacts before proceeding with a project.
Integrated Impact Assessment (IIA)	Combines different types of assessments (such as social, economic, and environmental) into a single process. It aims to provide a more holistic view of the impacts of a policy, program, or project.
Ecological Impact Assessment (ECIA)	Focuses specifically on the ecological impacts of a project or development. This includes assessing impacts on wildlife, habitats, and biodiversity.
Strategic Environmental Impact Assessment (SEIA)	A formalized, systematic, and comprehensive process of evaluating the environmental impacts of a policy, program, or plan, as well as its alternatives. It uses findings in publicly accountable decision-making.

MARCH 2023

The notification of Environmental Impact Assessment under the provision of Environmental (Protection) Act, 1986 was carried out for

1. River valley projects.
2. Development projects.
3. Tourism Projects.
4. Muncipal Waste Management.

ANSWER: B

Which of the following inter-governmental conferences recommended Environmental Education to cater to all ages and socio-professional groups in the population?

1. Stockholm (1972)
2. Tbilisi (1977)
3. Rio de Janeiro (1992)
4. Berlin (1995)

ANSWER: B

The inter-governmental conference that recommended Environmental Education to cater to all ages and socio-professional groups in the population was the **Tbilisi Conference (1977)**

Which one of the following fundamental duties makes sure that every citizen of India protects and improves the natural environment including forests, lakes, rivers and wildlife, and has compassion for living creatures?

1.51A (c)

2. 51A (e)

3. 51A (g)

4.51A (i)

ANSWER: C

Which one of the following is NOT a product-based tool of environmental preventive management?

1. Cleaner production

2. Eco - Labeling

3. Industrial Ecology

4. Life cycle assessment

ANSWER: A

Which of the following are NOT objectives of the Environment (Protection) Act, 1986?

A. Prevention of hazards to all living creatures (plants, animals and human) and property

B. Conservation of the natural heritage

C. Check on soil erosion and denundation in catchment areas in rivers, lakes and reservoirs

D. Maintenance of a harmonious relationship between humans and their environment

E. Increase in productivity of forests to meet the natural needs

Choose the correct answer from the options given below:

1. A, B and D only

2. B, C and E only

3. B, C and D only

4. C, D and E only

ANSWER: B

The objectives of the Environment (Protection) Act, 1986 primarily focus on protecting and improving the environment, as well as preventing hazards to all forms of life and property.

Given the provided options, the correct answer is:

2. B, C and E only

Here's a brief explanation:

- **B. Conservation of the natural heritage:** While important, this is not explicitly stated as an objective of the Act.
- **C. Check on soil erosion and denudation in catchment areas in rivers, lakes and reservoirs:** This specific objective is generally covered under the scope of other environmental acts and regulations.
- **E. Increase in productivity of forests to meet natural needs:** This objective falls under the ambit of the Forest Conservation Act and related policies, rather than the Environment (Protection) Act itself.

The Environment (Protection) Act, 1986 is broad in scope, focusing on environmental protection and the prevention of hazards, but it does not specifically address all these points.

Choose the correct statements

A. Basel convention was on transboundary movement of toxic waste.

B. Kyoto protocol was on banning of production of CFCs.

C. Montreal protocol was on controlling Global warming.

D. Earth Summit in the year 2002 was held in Johannesburg.

E. Convention on Biodiversity (CBD) was held in the year 1995.

Choose the correct answer from the options given below:

1. A, B, C and E only

2. A, B and C only

3. A and D only

4. B, C, D and E only

ANSWER: C

Let's review the statements one by one:

A. **Basel Convention**: This was indeed on the control of transboundary movements of hazardous wastes and their disposal. B. **Kyoto Protocol**: This was on reducing greenhouse gas emissions, not specifically on banning CFCs. C. **Montreal Protocol**: This was specifically on substances that deplete the ozone layer, including CFCs. D. **Earth Summit in 2002**: This was indeed held in Johannesburg. E. **Convention on Biodiversity (CBD)**: The CBD was actually held in 1992, not 1995.

Hence, the correct answer is:

3. A and D only

Following are the common procedural steps in ecological impact analysis process. Arrange them from the first to the last step.

A. Focussing

B. Impact Assessment

C. Impact Evaluation

D. Impact Mitigation

E. Scoping

Choose the correct answer from the options given below:

1. A, E, C,. B, D

2. E, A, B, D, C

3. E, B, C, D, A

4. A, B, C, D, E

ANSWER: B

The common procedural steps in the ecological impact analysis process, arranged from the first to the last, are:

E. Scoping → A. Focussing → B. Impact Assessment → C. Impact Evaluation → D. Impact Mitigation

So, the correct answer is:

2. E, A, B, D, C

Here's a brief overview of each step:

- **Scoping (E)**: Identifying the key issues and determining the focus of the impact analysis.
- **Focussing (A)**: Narrowing down the specific areas of concern and aspects to be studied in detail.
- **Impact Assessment (B)**: Analyzing the potential environmental impacts based on the identified focus areas.
- **Impact Evaluation (C)**: Evaluating the significance and severity of the identified impacts.
- **Impact Mitigation (D)**: Proposing measures to mitigate or manage the negative impacts.

Which of the following is related to 'MARRAKESH ACCORDS?

1. The Earth Summit (1992)
2. Berlin Mandate (1995)
3. Geneva Convention (1996)
4. Kyoto Protocol (1997)

ANSWER: D

4. Kyoto Protocol (1997)

The Marrakesh Accords were adopted in 2001 during the 7th Conference of the Parties (COP7) to the United Nations Framework Convention on Climate Change (UNFCCC) and provided detailed rules for implementing the Kyoto Protoco

SEPTEMBER 2022

Which section of Environmental (Protection) Act, 1986 deals with offences by companies?

1. Article 13
2. Article 14
3. Article 16
4. Article 18

ANSWER: C

Prevention, control, and abatement of environmental pollution

Section 7- Emission or discharge of environmental pollutants in excess of the standards

According to Section 7, it is prohibited for any person to discharge or emit any environmental pollutant in excess of the prescribed standards from any industry, operation, or process.

Section 8- Handling of hazardous substances

Section 8 lays down that all the persons handling any hazardous substances shall do so by complying with all the procedures and safeguards as may be prescribed.

Sample analysis

Section 12- Environmental laboratories

The Central Government is empowered to establish one or more environmental laboratories, or recognise any laboratory as an environmental laboratory to carry out the functions assigned under this Act. Rules regarding the functions, procedures, and other matters related to the environmental laboratory are to be framed by the Central Government by Notification in the Official Gazette.

Penal provisions under the Environment Protection Act

Section 15- General offences

Section 15 prescribes the penalty for general offences committed under this Act. if any person fails to comply with or contravenes any provisions of this Act, or rules made or orders or directions issued, he would be punishable with imprisonment for a term which may extend to five years or with a fine up to Rs. 1 Lakh, or with both. If the failure or contravention continues, then an additional fine which may extend to Rs. 5000 may be laid for every day the failure or contravention continues. And if this failure or contravention extends beyond one year after the date of convection, then the imprisonment can extend upto seven years.

Section 16- Offences by companies

For an offence committed by a company, Section 16 holds responsible the person who at the time the offence was committed was in charge of and responsible for the conduct of the company as well as the company. However, if it proved that any such person was liable exercised due diligence or that the offence was committed without his knowledge. Also, if it is proved that the offence was committed with the consent, connivance, or negligence of any director, manager, secretary, or another officer, then such person shall be liable to be proceeded against.

It is also specified that ‘company’ includes any body corporate, a firm, or any other association of individuals. The word ‘director’ also means ‘partner’ in relation to a firm.

Section 17- Offences by government departments

Section 17 lays down that for an offence committed by a government department, the Head of the Department shall be held responsible unless he proves that the offence was committed without his knowledge or that due diligence was exercised. However, if it is proved that the offence has been committed with the consent, connivance, or neglect of any officer other than the Head of the Department, then that officer shall be proceeded against and punished accordingly.

Other important provisions under the Environment Protection Act

Section 5A- Appeal to National Green Tribunal

Section 5A provides for the provision to appeal against an order or decision of the Appellate Authority under Section 31, by the aggrieved person. As per the Section, the appeal can be filed to the National Green Tribunal established under Section 3 of the National Green Tribunal Act, 2010.

Section 22- Bar of jurisdiction

Section 22 takes away the jurisdiction of civil courts from entertaining any suit or proceeding related to anything done or direction issued by the central government or an officer or authority in order to discharge the functions assigned by this Act.

"A financial mechanism through which developing countries would be rewarded for emission reduction achieved by taking action to reduce deforestation and forest degradation" refers to

1. Clean Development Mechanism.
2. Global Forest Carbon Mechanism.
3. Reduced Emission from Deforestation and Forest Degradation.
4. Incentive - Compatible Selection Mechanism.

ANSWER: B

Explanation:

Global Forest Carbon Mechanism:

- **The Global Forest Carbon Mechanism (GFCM) is an international initiative to reduce greenhouse gas (GHG) emissions from deforestation and forest degradation.**
- It encourages countries to invest in sustainable forest management practices that reduce GHG emissions while providing economic and social benefits.
- The GFCM is intended to provide financial incentives to countries that engage in forest carbon management activities and to direct private sector funds towards these activities. Through the GFCM, countries can earn carbon credits by protecting, restoring, and sustainably managing their forests. These credits can then be sold on the international carbon market to generate revenue for the country.

Hence, option 2 is the correct answer.

Additional Information

- **The Clean Development Mechanism (CDM)** is a United Nations-run carbon offset scheme allowing countries to fund greenhouse gas emissions-reducing projects in other countries and claim the saved emissions as part of their own efforts to meet international emissions targets. It is one of the three Flexible Mechanisms defined in the Kyoto Protocol.
- **Incentive- Compatible Selection Mechanism**: It ensures long-term and sustainable actions to minim global economic costs of achieving emission reduction of countries with targets. It aims to internalize the external values of environmental services. It mainly focuses on monitoring and enforcement of climate change framework with possible options for an incentive to capture and market the carbon values of forests.
- **REDD+** is a framework created by the UNFCCC Conference of the Parties (COP) to guide activities in the forest sector that reduces emissions from deforestation and forest degradation, as well as the sustainable management of forests and the conservation and enhancement of forest carbon stocks in developing countries. It aims at the implementation of activities by national governments to reduce human pressure on forests that result in greenhouse gas emissions at the national level, but as an interim measure also recognizes subnational implementation.

Which one of the following is correctly matched pair?

1. Biodiversity Conservation - Basel Convention
2. Global Warming - Montreal Protocol
3. Climate Change - Kyoto Protocol
4. Groundwater Pollution - Ramsar Convention

ANSWER: C

The correctly matched pair is:

3. Climate Change - Kyoto Protocol

Here's a brief explanation for each pair:

- **Biodiversity Conservation - Basel Convention**: Incorrect. The Basel Convention deals with the control of transboundary movements of hazardous wastes and their disposal.
- **Global Warming - Montreal Protocol**: Incorrect. The Montreal Protocol focuses on substances that deplete the ozone layer, not directly on global warming.
- **Climate Change - Kyoto Protocol**: Correct. The Kyoto Protocol specifically addresses the reduction of greenhouse gas emissions to combat climate change.
- **Groundwater Pollution - Ramsar Convention**: Incorrect. The Ramsar Convention is focused on the conservation and sustainable use of wetlands, not specifically on groundwater pollution.

Which among the following is the first major Environmental Protection act to be promulgated in India ?

1. Biological Diversity Act
2. Environmental Protection Act
3. Forest Conservation Act
4. Wildlife Protection Act

ANSWER: D

Which Article of the Constitution of India states that it shall be the duty of every citizen of India to protect and improve the natural environment and to have compassion for living creatures?

1. Article 48A
2. Article 51A
3. Article 21A
4. Article 13A

ANSWER: B

Which of the following principle was introduced in 1972 by the Organization for Economic Cooperation and Development (OECD) guiding principles concerning Institutional Economic Aspects of Environmental Pollution?

1. Frontier Diplomacy Principle
2. Polluter Pays Principle
3. Human Rights and Environment Principle
4. Principle of Environment Protection

ANSWER: B

The principle introduced in 1972 by the Organization for Economic Cooperation and Development (OECD) concerning Institutional Economic Aspects of Environmental Pollution is:

2. Polluter Pays Principle

The Polluter Pays Principle (PPP) stipulates that the polluter should bear the cost of pollution caused by them. It aims to internalize the external costs of production and consumption, ensuring that those responsible for pollution are held financially accountable for the environmental damage they cause.

Option	Explanation	Organization
1. Frontier Diplomacy Principle	This principle is not a recognized term in environmental policy or international law.	Not applicable
2. Polluter Pays Principle	The principle that those who produce pollution should bear the costs of managing it to prevent damage to human health or the environment.	Organization for Economic Cooperation and Development (OECD)
3. Human Rights and Environment Principle	This principle emphasizes the interconnection between human rights and a healthy environment, recognizing environmental protection as a fundamental human right.	United Nations (UN)
4. Principle of Environment Protection	This principle broadly refers to various measures and approaches aimed at protecting the environment, but it is not a specific, formalized principle like the Polluter Pays Principle.	Various international organizations and national governments